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Perspective

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Positive tipping points for nature

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A radical acceleration of change is needed to reach widely agreed nature-positive goals for 2030 and beyond. Triggering positive tipping points where strong amplifying feedback supports self-propelling nature-positive change can help achieve this acceleration. These include ecological, social–ecological and social tipping points that abruptly regenerate ecosystems and resources, spread nature-positive activities and reduce key drivers of nature loss. Together they have the potential to positively tip large-scale recovery of nature. Affecting key leverage points by facilitating online collective learning, properly valuing nature in economics and tipping worldviews to ecocentrism would further enable positive tipping points for nature.

There is widespread policy support for humanity to become 'nature positive' by 2030—meaning a net global gain in biodiversity indicators relative to a 2020 baseline[1]. The Kunming–Montreal Global Biodiversity Framework enshrines 2030 targets including the restoration of 30% of all degraded ecosystems and conserving 30% of land, waters and seas. Yet over halfway through the United Nations Decade on Ecosystem Restoration progress is far slower than it needs to be to meet these goals. A major acceleration of change is required, with a range of actions proposed to help achieve this[2,3]. This Perspective highlights how identifying and triggering positive tipping points can make a key contribution to achieving the required acceleration (Fig. 1).

Tipping points happen at a threshold when strongly amplifying feedback supports self-propelling change within a system towards a qualitatively different state[4]. The resulting hard-to-reverse change is sometimes called a 'regime shift'[5] or 'critical transition'[6]. Much current focus is understandably on the risk of 'negative' (undesirable) tipping points in some ecosystems (for example, coral reefs[4]), social–ecological systems[5] (for example, fisheries) or biomes (for example, the Amazon rainforest[4]). However, if nature can be negatively tipped then it can also (in principle) be positively tipped (back) into a more desirable state[7]. Success stories include the reintroduction of wolves to Yellowstone National Park[8], the restoration of eutrophic lakes[9] or the abrupt recovery of collapsed fish stocks (and associated fisheries)[10]. Success can involve tipping points in social norms towards cooperation[11]. Equally, the spread of nature-positive activities such as conservation initiatives[12,13] can reach a 'critical mass' tipping point of self-propelling adoption. Furthermore, wider social norms of consumption can (in principle) tip in a direction that reduces direct drivers of nature loss—for example, shifts away from plastic littering or meat-eating[14]. Thus, positive tipping points for nature encompass ones that occur in nature and ones that occur in social systems.

This Perspective synthesizes examples of, and evidence for, positive tipping mechanisms in nature, societies and their coupled dynamics (Table 1). It shows that by deliberately activating these mechanisms, self-accelerating nature-positive change can be triggered, across a range of systems and scales. This complements recent literature identifying positive tipping points in social–technological systems—especially the energy system—to tackle climate change[4,15]. They can reduce a key driver of loss of nature, while positive tipping points for nature can benefit climate through carbon removal and enhancing evapotranspiration. There are, however, important conceptual distinctions between them. First, nature already has its own built-in feedback loops—absent humans—that we can learn to work with to positively tip change, whereas human activities are always part of social–technological feedback loops. Second, while positive tipping points in technology adoption can ultimately spread worldwide[15], positive tipping points of nature are specific to a climate and ecological context. In the following, I define key concepts. Then I synthesize the literature to identify types of positive tipping point for nature and actions that can trigger them, proceeding from ecological to social tipping and from smaller to larger scales, and illustrating with diverse examples. I close with a forward look including three generic changes that would help to enable positive tipping points for nature.

Key concepts

Tipping points

A tipping point is "when change in (part of) a system becomes self-perpetuating beyond a threshold, leading to substantial, widespread, frequently abrupt and often irreversible impact"[4] (Fig. 1). It typically takes some change in the drivers (forcing) of a system to bring it to a tipping point (Fig. 2a). Before a tipping point, balancing (negative) feedback maintains the initial state of the system, but this gets weaker

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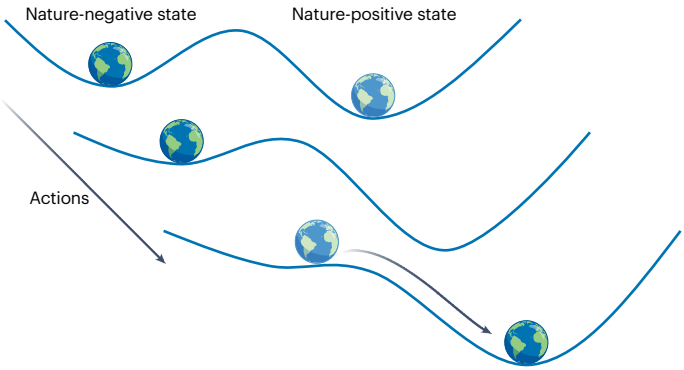


Fig. 1 | Simple representation of a positive tipping for nature. The valleys represent alternative stable states of a system (be it ecological, social or social–ecological), which are maintained by balancing (negative) feedback mechanisms. As actions force the system towards a tipping point, balancing feedback gets weaker (the left valley gets shallower), until at the tipping point (where the valley disappears), reinforcing (positive) feedback within the system gets strong enough to support self-propelling change into the alternative system state (the ball rolling down the slope). That new state is stabilized by different balancing feedback mechanisms.

typically stabilize the new state. This creates hysteresis—the state of the system depends on its history—meaning that in a system with, for example, two alternative stable states there are different tipping points for the two directions of transition (Fig. 2a).

Tipping points can occur in many types of complex system, including ecosystems, social systems and social–ecological systems in which people and nature are inextricably linked (such that both the social and ecological components exert strong influence over the outcomes). Here I distinguish between ecological tipping points that are triggered by human activities (for example, reintroduction of wolves to Yellowstone[8]), social tipping points that can cause ecological change[16] (for example, self-propelling diffusion of conservation initiatives[12,13]) and social–ecological tipping points that arise only because of the coupling of social and ecological dynamics (for example, coupled abrupt shifts in resource harvesting norms and resource levels[17]).

Whether a system can tip is determined by whether reinforcing (positive) feedback can get strong enough to (temporarily) support self-propelling change. If not (Fig. 2b), reinforcing feedback may still amplify change in a constrained way but does not produce alternative stable states of the system. Hence, to identify the potential for tipping point behaviour and distinguish it from non-tipping forms of (environmental and/or social) change, it is important to establish the existence and strength of reinforcing (positive) feedback(s)

Table 1 | Types of positive tipping point for nature

Type	Description	Examples	Actions/interventions
Trophic cascade	Reintroducing a trophic level triggers a cascading change in species abundance across trophic levels	Yellowstone wolves, Pacific sea otters, piscivorous fish in shallow lakes, shorebirds in salt marshes, herbivorous fish in coral reefs	Keystone species reintroduction or reduced harvesting, removal of antagonistic species
Biogeochemical tipping	Reducing driver(s) past a threshold triggers self-propelling change in biogeochemical conditions	Recovery from eutrophication of shallow lakes, coastal zones, coral reefs	Reduce biogeochemical driver(s), for example, nutrient loading of aquatic systems
Biophysical tipping	Reaching a critical density of ecosystem engineers triggers self-propelling change in physical conditions	Recovery of mangroves, salt marshes, mussel beds, oyster reefs, seagrass meadows, *Sphagnum* bogs, arid vegetation, coral reefs	Clumped reintroduction of ecosystem engineers, assisted by physical structures, biomimicry of key traits
Population tipping (Allee effect)	Reaching a critical density of breeding organisms triggers self-propelling population recovery	Recovery of fish stocks, coral reef communities	Enforce maximum sustainable yield of fish, marine protected areas, reintroducing coral larvae
Cooperation over common-pool resource	Reaching a critical mass of adopters of a cooperation norm triggers self-propelling adoption and resource recovery	Fisheries management, groundwater management	Community governance of common-pool resource, enforcement
Escape from resource poverty trap	Regenerating a resource base reaches a critical level where benefits support self-propelling growth	Organic farming, System of Rice Intensification	Regenerate soil fertility, reduce nutrient input, reduce water input, training, access to labour at key times
Diffusion of nature-positive initiatives	Reaching a critical mass of adoption of an initiative triggers self-propelling adoption	Forest management, marine reserves, groundwater management, agroforestry, organic farming	Community ownership, training, access to resources, supportive bylaws, market formation, incentives
Shift in consumption norms	Reaching a critical mass of adopters of a new norm triggers self-propelling adoption	(Prospective) dietary change to reduced meat consumption	Change dietary guidelines, choice architecture, shift subsidies, taxation, control advertising
Shift in individual worldviews	Cognitive dissonance reaches a critical point where worldview tips to a new coherent state	(Prospective) shift to ecocentrism including rights of nature	Ecological education, access to information on consequences of anthropocentrism

as the tipping point is approached. At the tipping point, reinforcing (positive) feedback within the system gets strong enough to overwhelm balancing feedback and support self-propelling and hard-to-reverse change towards a new system state. Initially this produces accelerating change, which then reaches a maximum rate before slowing down as the system approaches its new state. The maximum rate of change is governed by the timescales of the system and is not always rapid to human perception. Different balancing feedback ('lock-in') mechanisms within a system. If a system can tip, actions are usually required to bring it to a tipping point (Figs. 1 and 2a). The amount of action required depends on how far the system is from the tipping point. Relevant action can seek to alter the drivers of a system (Fig. 2a), to directly alter the state of the system, to strengthen reinforcing feedback(s) in the system and/or to weaken damping feedback(s) that maintain the initial state. A range of actors can be involved in actions to bring forward positive tipping, including the inhabitants

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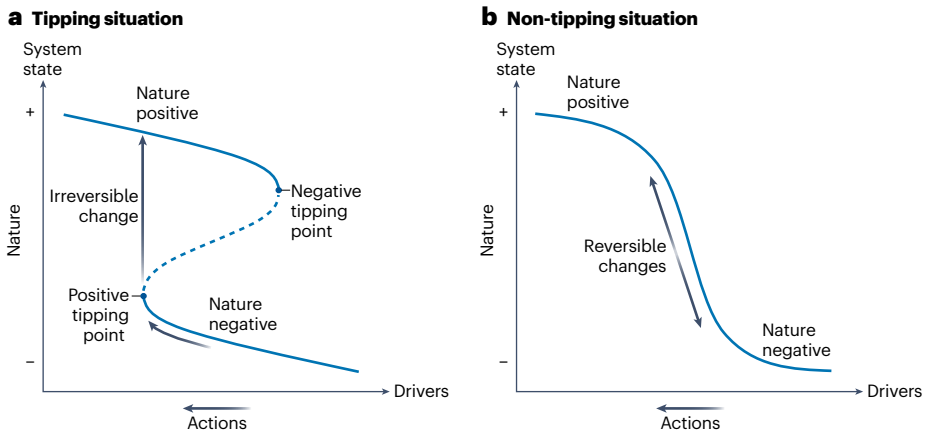


Fig. 2 | Tipping and non-tipping situations. a, Schematic representation of a system with tipping points between nature-positive and nature-negative stable states (solid lines, equivalent to the valleys in Fig. 1) separated by an unstable steady state (dashed line, equivalent to the hilltop in Fig. 1). Changes in drivers can change the stability of these states and stability is lost for each state at its tipping point. Actions to reduce drivers are shown as one way of triggering a positive tipping point leading to irreversible change into a nature-positive state (equivalent to the depiction in Fig. 1). **b**, Schematic representation of a system without tipping points where changing drivers causes changes in its state. Here, actions to reduce drivers can make the system's state more nature positive, but if those actions are reversed, the change in system state will also be reversed.

of a social–ecological system, conservationists, policymakers and financers. Their actions may be resisted by other actors who perceive they benefit from maintaining the status quo. However, if tipping occurs, some actions (for example, policies) that were required to trigger it may not need to be maintained as the resulting change has become hard to reverse (Fig. 2a). In contrast, in non-tipping situations (Fig. 2b), temporary actions are rarely sufficient to maintain change as their effects are reversible.

Defining positive tipping for nature

'Positive' for nature can be taken in a mathematical sense to mean that passing a tipping point leads to net gain of nature in going from one system state to another (whether the system is ecological or social). Numerous measures of 'nature' could be used to assess this. The original 'nature positive' proposal[1] suggests "species distribution, abundance, functional traits, genetic diversity, demographic trends", "the intactness and integrity of ecosystems and biomes" and "the functioning of ecological and global processes such as hydrology, rainfall patterns and migration that support biodiversity". Productivity and biomass are further candidates given human appropriation of net primary productivity and the massive loss of vegetation, soil and wild animal biomass that have occurred due to human activities. However, more is not always better locally, as when increased productivity due to eutrophication creates aquatic dead zones[18]. Accepting that there is no single universal metric, and that some context-specific (and partly subjective) judgement is necessary, ecologists tend to agree on what counts as net gain of nature.

There is a further subtlety for ecosystems with tipping points that have alternative stable states under their current boundary conditions—such as large areas of the tropics that can be forested or savanna[19]. Both states may be considered 'natural' and the observed state is partly a function of history (path dependency), so how does one judge which state is (more) 'positive'? The above metrics could be invoked, or systemic qualities such as which state is more stable (for example, has a larger basin of attraction), but the latter may change with changing drivers, for example, climate change[19]. In practice, historical familiarity tends to factor into the judgement and the state prior to intense anthropogenic disturbance is usually deemed preferable—as in conservation more broadly. Accepting this subtlety, ecologists are usually able to agree when an ecosystem is in a 'degraded' state and what counts as a 'regenerated' (nature positive) state.

Clearly 'positive' for nature is also often taken in a normative sense—of being a value judgement of what is better for nature. Giving nature's needs some priority over human needs is consistent with an eco-centric worldview and ethics (also termed interspecies justice); that humans are part of nature, depend upon nature and that nature has intrinsic (moral) value[20]. Given that humans cannot exist without the rest of nature, this ethical position is also 'people positive' in a collective, existential sense—we can only survive and flourish if the rest of nature does. It can thus also fit with a more anthropocentric ethic, including an instrumental valuing of nature, which recognizes numerous material and immaterial benefits ('ecosystem services') flowing from nature to people. These include benefits of nature to physical and mental health, including promoting awe and other positive emotions[21].

Such benefits to humans do not define what is positive for nature. However, for change to happen, considering who benefits from and who may be harmed by specific tipping points is an important practical (as well as ethical) consideration. Nature-positive change can align with tackling inter- and intra-generational justice issues[2], for example, by increasing the biosphere's ability to provide for the needs of future generations. Furthermore, Indigenous peoples with eco-centric worldviews who have suffered injustices are often best placed to regenerate nature[20]. However, historically, many conservation initiatives have excluded already marginalized peoples and thus accentuated human injustices[22]. To achieve a nature-positive tipping point for a social–ecological system it is helpful if its human inhabitants perceive it will be positive for them. This may be aided by privileging the inhabitants' ethical position over those who live elsewhere, but the inhabitants can still disagree about what is positive for them (as explored further below).

I now turn to identifying types (Table 1) and examples (Fig. 3) of positive tipping points for nature.

Positively tipping ecosystem recovery

The same mechanisms that underlie negative ecological tipping points can be leveraged to positively tip ecosystems back from a degraded state[23]. Hysteresis means more effort is usually required to reverse drivers to tip recovery of a preferred ecosystem state than triggered collapse (Fig. 2a), and some drivers—notably climate change—are harder to reverse than others. Time lags in societal responses can further inhibit recovery. Nevertheless, by altering drivers and working with feedbacks, in threshold-based management[24], numerous degraded ecosystems

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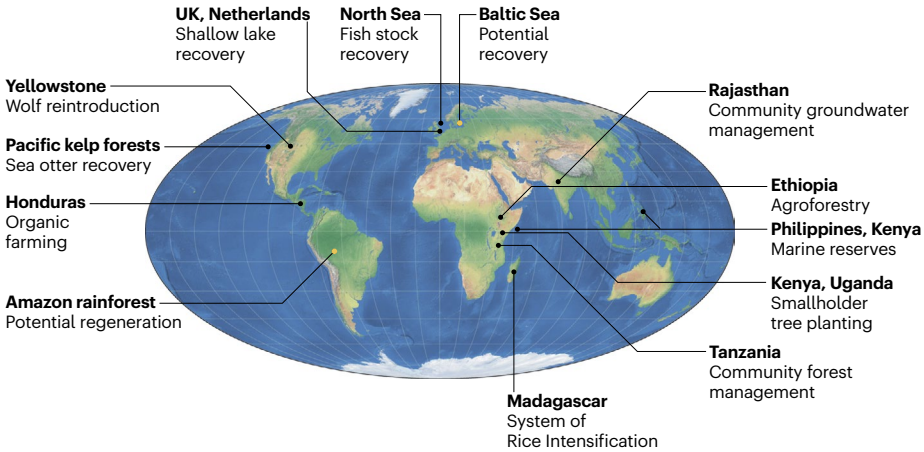


Fig. 3 | Location of examples of positive tipping points for nature. Black circles show early locations of initiatives that have often diffused more widely. Yellow circles indicate two larger-scale systems with the potential to be positively tipped (Box 1). Credit: basemap, Tobias Jung under a Creative Commons license CC BY-SA 4.0.

BOX 1

Two examples of potential large-scale positive tipping of nature

The Baltic Sea, along with its catchments, has experienced multiple negative ecological tipping points[93], but it has the potential for large-scale positive tipping of recovery. This shallow, brackish coastal shelf sea is effectively a giant estuarine system. Historically it has seen a trophic cascade tipping point in the 1950s–1960s with the removal of seals as top predators. Then the 1980s saw a pelagic tipping point from overfishing, deterioration of spawning grounds and climate change, with an abrupt switch from cod to sprat and herring. Numerous coastal tipping points have occurred involving the loss of macrophytes (seagrasses and perennial algae) through trophic cascades. There has also been a tenfold increase in deoxygenated dead zones over the past century, driven by nutrient runoff and eutrophication and reinforced by phosphorus recycling from sediments. Some nature-positive change has already happened through the return of top predators, coordinated fisheries management and reductions in nutrient loading. But these changes have suffered systematic delays and been thus far insufficient to positively tip away from eutrophication and dead zones[94]. Achieving a larger-scale positive tipping point for nature would benefit from positive tipping in diets, reducing the demand for intensive agriculture on land and associated nutrient loading (as well as demand for fish). Already Denmark, for example, has revised dietary guidelines and is implementing new agricultural taxes. Reducing nutrient loading also requires institutional change in the Common Agricultural Policy and its enforcement. This would benefit from a coordination tipping point among multiple stakeholders including policymakers, farmers, environmental groups and consumers[95].

The Amazon rainforest is recognized as a tipping element in the climate system that could be negatively tipped to a degraded forest or savanna state at large scales, through a mix of climate change and deforestation drivers and biophysical amplifying feedback. The Amazon has already lost ~20% of forest cover, and 38% of the remaining forest is degraded[96], much of it by climate change. Demand for beef, much of it for Brazilian consumption[97], remains the single largest driver of deforestation. Hence a positive tipping point in diets towards reduced beef consumption, encouraged by targeted tax reform and improved traceability[97], could play a critical role in stopping deforestation. Then planting and natural regeneration have the potential to positively tip large-scale forest recovery. Many regions where forest has been removed, predominantly in the southern Amazon, exhibit bistability of tree cover[19]. Corresponding tipping points arise locally from forest geometry and fire feedback[98], and extend to larger scales through moisture recycling transporting water downwind[99], and through the release of latent heat supporting the monsoonal inflow of moist air[100]. Understanding of these effects can be leveraged to design interventions that maximize the potential to trigger propagating positive tipping. For example, reforestation efforts can consider localized geometry to maximize the resulting destabilization of the grassland/pasture state[98]. Regions can be targeted for regeneration based on their contribution to moisture recycling and wind patterns to maximize longer-range benefits[99,100]. Such spatially coordinated action would benefit from positively tipping cooperation among relevant actors.

have been positively tipped into a regenerated state. This can leverage a mix of trophic, biogeochemical and biophysical reinforcing feedback mechanisms (Fig. 4).

Trophic feedbacks can be strong enough that the removal or reintroduction of one trophic level (for example, predators or large herbivores) triggers a cascading change in species abundance across trophic levels. Widespread removal of apex predators by humans means there is considerable scope for their reintroduction to positively tip trophic cascades that suppress herbivores and abruptly restore vegetation[25]. For example, the deliberate removal of wolves from Yellowstone National Park in the early twentieth century increased the abundance of elk and suppressed several plant species that the elk browse upon including aspen, cottonwoods and willows. The reintroduction of wolves in 1995–1996 led to a positive tipping point of riparian vegetation recovery[8] and boosted the numbers of scavengers, songbirds, bison and beaver—although the strength of the trophic cascade may be overestimated due to sampling bias[26]. A further, aquatic example is the recovery of sea otters in Pacific kelp forests[24]. Their removal

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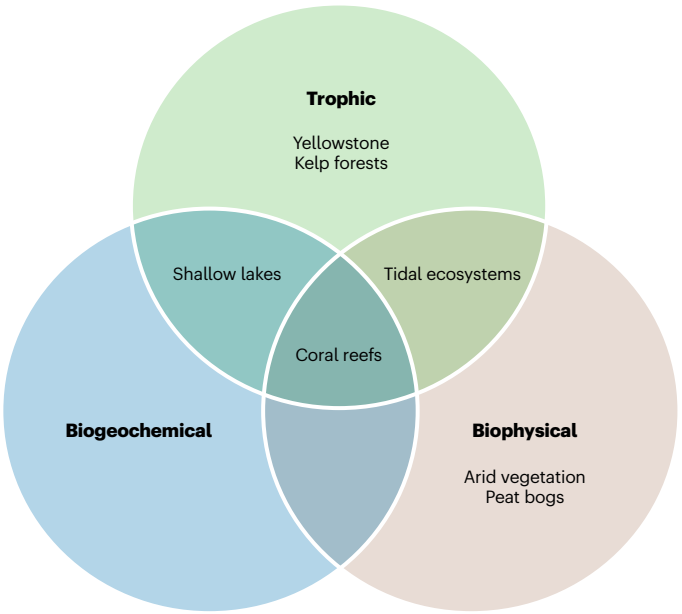


Fig. 4 | Types of reinforcing feedback mechanism involved in positive tipping of ecosystem recovery. Positioning of example ecosystems indicates which types of reinforcing feedback are involved in their tipping points (see also Table 1).

caused sea urchin populations to escalate and kelp populations to collapse. Recovery of sea otter populations (or their reintroduction, for example, in Alaska) has tipped kelp forest recovery, but due to hysteresis required a greater otter density than that at which collapse occurred[24]. It has benefited biodiversity, fin fisheries, carbon storage, recreation, tourism and coastal protection, but in places that support lucrative urchin and shellfish fisheries, recovery is resisted by some[24]. Such 'trophic rewilding'[27] has much wider potential to positively tip ecosystem recovery.

Biogeochemical amplifying feedback can get strong enough to support tipping points in the nutrient status of aquatic ecosystems. Notably, numerous shallow lakes have been negatively tipped by phosphorus loading, promoting productivity, anoxia, and phosphorus recycling from sediments that amplified change. This is often further reinforced by trophic cascades, whereby increasing planktivorous and benthivorous fish cause loss of zooplankton and benthic filter feeders[28]. Despite hysteresis, efforts starting in the Norfolk Broads and the Netherlands have positively tipped lakes back into a clear water state by reducing external phosphorus loading and by activating reinforcing feedbacks, including introducing filter-feeders such as zebra mussels, and piscivorous fish that trigger reverse trophic cascades[9].

Biophysical amplifying feedback can get strong enough to support tipping points in highly stressed tidal ecosystems including mangroves, salt marshes, mussel beds, oyster reefs and seagrass meadows[29]. In these systems, ecosystem engineers stabilize sediments and attenuate waves and currents, facilitating their persistence and spread, sometimes aided by producing spatial patterns[30], or by altering redox or salinity state. Around 35% of mangroves, 40% of salt marshes, 85% of oyster reefs and 30% of seagrass have been negatively tipped[29]. However, positive tipping of recovery can be triggered by leveraging reinforcing feedbacks through clumped reintroduction, for example, of salt marsh plants[31], aided by simple physical interventions like fencing for mussel beds[32] or permeable dams for mangroves[33], or with more sophisticated mimicry of biophysical traits with engineered structures[29,34]. Positive tipping can be further encouraged by physically excluding bioturbating animals that destabilize sediments, for example, in seagrass meadows[35], or by leveraging trophic cascades, for example, by encouraging shorebirds that suppress grazers and bioturbators in salt marshes[36]. The recovery of other stressed ecosystems, such as arid vegetation and peat bogs, can also be positively tipped[29].

In warm-water coral reefs, a combination of trophic, biogeochemical and biophysical amplifying feedback mechanisms can together get strong enough to create tipping points. Many reefs have been negatively tipped into an alternative macroalgal-dominated state by nutrient loading and overfishing of species (for example, parrotfish) that consume macroalgae, or by severe disturbances including storm destruction, attacks of crown-of-thorns starfish and marine heatwaves triggering mass bleaching events. However, recovery is possible if herbivorous fish are abundant enough and nutrient levels low enough to prevent macroalgal proliferation[37,38]. Many reefs may now have passed a negative tipping point due to ongoing global warming[4]. But where local drivers also contribute to tipping, reef recovery may still be positively tipped by reducing those drivers, including controlling fishing[39] and anthropogenic nutrient loading[40]. Positive tipping can be further aided by activating reinforcing feedbacks, including deliberately introducing juvenile corals[41] or artificial reef structures[42], supporting seabirds and their nutrient supply[43], and (possibly) through assisted evolution[44]. Natural variability, such as persistent cloudy conditions that impair macroalgal production[40], can open a window for positive tipping.

Positively tipping social–ecological systems

Warm-water coral reefs, bivalve beds and lakes often support fisheries or aquaculture, making them also social–ecological systems. Such common-pool resource systems can exhibit tipping points[10], including ecological tipping points triggered by human activities, social tipping points that transform ecology or coupled social–ecological tipping points[17]. The same reinforcing feedback mechanisms that underlie negative tipping points of collapse of common-pool resources can be leveraged to positively tip recovery.

Careful management can positively tip the recovery of fish stocks that have collapsed due to overfishing. Collapse has happened to cod stocks in many locations and to multiple species in the North Sea[45], sometimes exacerbated by fishers working harder to maintain the same catch as populations are declining[46]. Strong amplifying feedback between population density and reproductive success (known as an Allee effect) produces a tipping point at a critical density of breeding fish[47]. A positive tipping point can be triggered by reducing fishing activity (the driver) such that spawning stock biomass and recruitment of juvenile fish recovers to the critical density. In pelagic fisheries, positive tipping can be triggered by enforcing a maximum sustainable yield—the highest yield that can be taken without markedly affecting reproduction. This typically requires short-term reduction in fishing effort, which in turn requires strong regulatory enforcement and punishment of those who do not comply. Enforcement of a maximum sustainable yield has produced positive tipping points for recovery of plaice and hake stocks in the North Sea[45]. However, cod stocks in multiple regions have thus far failed to recover[45,48], because there is additional amplifying feedback generating hysteresis, whereby increased sprat stocks outcompete larval cod for their zooplankton prey[48].

For coastal fisheries, marine protected areas (MPAs) can help positively tip fish stock recovery, by providing safe spawning areas and 'spillover' of fish into the surrounding waters. MPAs can be particularly effective on coral reefs as these naturally support transient fish spawning aggregations, although fishers often initially object to the loss of fishing grounds[49]. An early example of positive tipping was the formation of a marine reserve on Apo Island in the Philippines in 1982, which reversed a vicious cycle of overfishing and supported recovery of a sustainable fishery, where fishing effort declined over time, improving livelihoods and increasing tourism and community wellbeing[10,50]. A Kenyan coral reef marine park triggered similarly positive results[51]. MPAs may also help to positively tip the recovery of whole coral reef ecosystems as they can stop coral loss and old, well-enforced, no-take MPAs can benefit coral cover and depress algal cover[52]. MPAs have

limited effects on reef recovery from severe shocks[37], but modelling suggests that MPAs could positively tip reef recovery by reducing fishing of herbivorous fish that consume macroalgae[53] aided by raising social awareness of coral reef endangerment[54].

Managing fish stocks is an example of cooperating to manage a common-pool resource. Such cooperation is the evolved social norm in many cultures[11] (meaning a predominant behavioural pattern within a group, supported by a shared understanding of acceptable actions and sustained through social interactions within that group[11,14]). However, the norm of cooperation can break down into an alternative 'tragedy of the commons' state, with tipping points between these states. When the common-pool resource also has a tipping point, as in fisheries, uncertainty or not about where that lies can determine whether social dynamics tip to cooperation or not[55]. But even when a common-pool resource lacks a tipping point, there can be a positive tipping point to social cooperation that produces nature-positive change.

Groundwater management provides an example of a cooperation-driven positive tipping point, which is predicted to be sensitive to cultural values, monitoring and enforcement[56]. Positive tipping has been seen in drier parts of India, where traditional practice was for communities to cooperate to maintain artificial reservoirs that capture the monsoon rains, which then infiltrate to recharge groundwater that in turn supports vegetation, agriculture and the community[57]. With independence, the state appropriated control of water resources, and in the Alwar district, deforestation led to the reservoirs (johadi) silting up. Villagers shifted to digging ever-deeper wells, but water supplies dwindled and with them crops and livestock. Forests were further exploited, women had to walk longer distances to get water and men left to work elsewhere. Then in 1984, in Bhikampura village, a collaboration between the villagers and youth organization Tarun Bharat Sangh started by reinstating one johad. Rapid positive results triggered reinforcing feedback of increasing cooperation to restore more johadi. Agricultural cropping increased, forests recovered, women's workload declined, men returned. Cooperative village-level governance of water resources resumed through traditional Gram Sabha (governance fora)—a positive tipping point to cooperation to manage a common-pool resource.

The coupling of social and ecological systems can also produce a (social–ecological) tipping point in both, even when neither would independently exhibit one[17]. Resource-degradation poverty traps are an example seen in many developing countries, where households and their supporting ecological resources, such as soil or cattle, get stuck in a deleterious state, often driven by forces beyond their control[58]. A common feature of the 'trap' is that the people involved lack the means, derived from their resource base, to improve that resource base and instead continue to deplete it. An alternative stable state exists for those with greater means derived from greater resources to nurture and grow that resource base, in a reinforcing feedback of increasing yields, profit and reinvestment. Positively tipping out of poverty traps into a better state for both nature and people is therefore possible, although successfully identifying how has proved difficult in many cases[58].

Positive tipping can involve regenerating soil and its fertility[58], which benefits the soil biota and surrounding vegetation as well as agriculture. For smallholder farmers in steeply sloped parts of Honduras, soils are often heavily eroded, and their fertility depleted[59]. Adopting organic farming utilizing animal manure can positively tip households out of the resource-degradation poverty trap, as it has low cost and can help retain soils, boost their fertility, increase yields and thus boost financial returns, which are further enhanced if the farmers can access organic markets, where the produce attracts a premium[59]. For smallholder rice farmers in Madagascar, conventional practices have generated widespread forest loss, soil erosion, soil nutrient loss, poor yields and farmers trapped in poverty[58]. Adopting the System of Rice Intensification (SRI) can positively tip farmers out of the poverty trap as it can achieve high yields with low nutrient and reduced water inputs[60], boosting income.

Positively tipping diffusion of initiatives

The social spread of nature-positive initiatives can also exhibit a positive tipping point of self-propelling adoption. This includes the spread of some of the examples above of positive tipping points in ecosystems and social–ecological systems.

The theory of diffusion of innovations[61] focuses on how the more people who adopt something new, the more others they can collectively influence to adopt, producing reinforcing feedback. Adoption can spread through learning directly from one another, increasing availability of required tools or expertise, or changing social norms that others conform to. The strength of amplification also depends on the attractiveness of adoption. If amplification is strong, it can support immediate self-propelling adoption, but usually there is some resistance to adoption. This may be overcome at a critical mass tipping point, where one more person adopting triggers self-propelling adoption in a population[62]. If adoption becomes more attractive the more who adopt (for example, because costs decline and/or benefits increase) that also produces reinforcing feedback, which can generate a critical mass tipping point. So too can network effects whereby adoption increases benefits for those who have already adopted (as well as those yet to adopt). If a critical mass tipping point exists, it is usually early in an overall 'S-curve' pattern of uptake. However, not all diffusion involves a tipping point, and in some cases diffusion can stall at a low level of adoption, or there can be abandonment (that is, reversibility).

Potential cases of positive tipping include rapid, widespread diffusion of some conservation initiatives[12,13]. For example, success on Apo Island inspired a diffusion of marine reserves in the Philippines that was dominated by the reinforcing feedback of social learning[63]. Locally managed marine (protected) areas first established in Fiji have also diffused widely within and across Pacific islands[12]. Here, benefits to local people, including autonomy, can determine whether reinforcing feedback gets strong enough to tip self-propelling adoption. Success in Fiji was linked to reorganizing resource management around customary tenure systems. In contrast, Samoa had government incentives for aquaculture and fishing subsidies, but when that investment declined so did adoption, indicating lack of a tipping point. The scaling of different conservation initiatives in Tanzania has also depended on benefits to local villagers, including autonomy, with community-based forest management having the greatest uptake[12]. Modelling can forecast where there is greatest potential for rapid scaling of adoption[63] and social network analysis can identify key players[64] who can facilitate positive tipping.

Sustainable water management and agriculture practices highlighted above have also shown evidence of self-propelling adoption. In Rajasthan, more than 3,000 johadi have now been reinstated by communities[57], and community-based groundwater management, which improves productivity by up to 250%[65], has spread more widely across Asia. At the aquifer scale, spread can be amplified by network effects whereby individual benefits (groundwater supply) increase with others' adoption. Successful diffusion of organic farming practices in Honduras shows other reinforcing feedbacks, from availability of information increasing with others' adoption, and from conformity to norms[59]. The SRI has spread to over 50 countries and diffusion within Madagascar has been amplified by training being passed on[66]. However, the SRI is labour-intensive at a time of year when liquidity is low and labour effort is already high, making it hard for those trapped in poverty to access the benefits[66]. Consequently, adoption in Madagascar has been slow, and abandonment high[66], indicating lack of a tipping point.

Other sustainable agriculture practices have exhibited self-propelling adoption, including agroforestry initiatives among smallholder farmers in the tropics[67]. The International Small Group and Tree Planting initiative (TIST) originating in Tanzania has spread rapidly in Kenya and Uganda[68] (and to India) aided by a structure designed to

maximize autonomy and social learning, and by providing multiple benefits to adopters, including carbon payments, helping alleviate household credit constraints, economically empowering women and greening the surrounding landscape[69]. In developed countries, highly variable adoption of organic farming[70,71], for example, 25% in Austria but only 2% in the Netherlands, indicates resistance to adoption that must be overcome to reach a positive tipping point[71]. Government-set goals and initiatives, market formation and resource mobilization (for example, subsidies and incentives) can help enable positive tipping[71].

The spread of nature-positive initiatives can also be propelled by ecological and biophysical reinforcing feedback mechanisms. Reintroduced keystone species can replicate and disperse (like all species), thus amplifying their ecological effects. Evapotranspiration from regenerating vegetation can support precipitation and thus vegetation growth downwind. Such moisture recycling[72] is particularly important in, for example, the western Amazon and La Plata basins[73], the Congo basin and agricultural areas of sub-Saharan Africa[74]. This suggests regions where revegetation efforts would promote their own spread (see Box 1 for an example).

Positively tipping consumption behaviour

Finally, there is the potential for positive tipping points in patterns of consumption to reduce key drivers of nature loss. Notably, habitat loss to agricultural expansion is the most important driver of nature loss, in turn driven primarily by dietary trends towards more meat consumption[75]. However, against this global average trend, in several rich nations that overconsume meat, there have been recent reductions in meat consumption, which is decoupling from economic growth[76].

Dietary choices are strongly influenced by social norms, and subject to other reinforcing feedbacks, suggesting the possibility of tipping points in the diffusion of lower-meat diets. At a personal level, a positive experience of eating plant-based protein encourages further consumption[77]. At a social level, more plant-based meals are consumed by those whose family and friends exhibit the same preference[78], seeing others consume a more sustainable option in a restaurant encourages uptake[79], and doubling the proportion of vegetarian meals on a menu increases vegetarian sales by 41–79%[80]. There are increasing returns to adoption as rising investment in alternative proteins is improving the quality and diversity of meat-free options and driving down their costs. There can also be reinforcing policy feedback as politically acceptable policies such as changing dietary guidelines can raise awareness and start to shift social norms and behaviours, thus increasing public support for more stringent policies such as taxes, which have proved effective at shifting dietary norms[81].

That said, strong balancing feedbacks are opposing dietary change. The meat and dairy industry spend 190 times more on lobbying than the alternative protein industry in the USA (and three times more in the European Union (EU))[82]. In the EU, four times as much farming subsidy goes into animal products as plant ones, EU cattle farmers get over half of their income directly from subsidy, and the EU spends almost twice as much promoting meat and dairy products as fruit and vegetables. Public investment in alternatives is low: £33 million invested in alternative proteins in the EU and USA combined for 2014–2020, versus £35 billion supporting meat and dairy production[82]. UK spending on a flagship healthy- eating campaign (£5.2 million per year) is dwarfed by annual advertising of unhealthy products (£143 million per year). Several governments have rejected scientific recommendations that would support changing diets, indicating that livestock products retain strong cultural value in many countries.

A framing of dietary change as good for both planetary and personal health, and essential to achieving 'nature positive', could help shift collective understanding and norms. In the 2020 United Nations Development Programme Peoples' Climate Vote, 'conservation of forests and land' was the most popular of 18 climate policy suggestions (with 54% average support), yet 'promote plant-based diets' was least popular (30% average support)—despite being the most effective way of conserving forests.

Looking forward

This synthesis of evidence from ecosystems, societies and their coupled dynamics shows how positive tipping points can support self-propelling nature- positive change across a range of systems and scales. Box 1 offers two prospective examples of how the tipping mechanisms outlined could be jointly leveraged to positively tip larger-scale natural systems.

A key research opportunity is to test whether resilience-sensing methods can indicate the potential for systems to be positively tipped[83]. These methods use direct measures of recovery rate from perturbations, or statistical time series indicators of this, to deduce the resilience of the current state of a system (where slow recovery indicates low resilience). Resilience declines on approaching a tipping point. Finding that an undesirable state lacks resilience or is losing resilience could therefore provide an 'early opportunity signal'[83,84] that it can be tipped out of, inspiring actors to intervene. Direct and remotely sensed data can reveal the resilience of degraded ecosystem states[85], and social data can be used to sense the resilience of incumbent norms or behaviours[83]. Alongside indicating when interventions will be most effective, resilience sensing may indicate how much effort will be needed to achieve positive tipping, and whether interventions are having the desired effect (of destabilizing the incumbent state) or not.

A corresponding theory of change requires identifying what actions by which actors can create enabling conditions and help trigger specific positive tipping points[84], who opposes such actions, and the resulting feedbacks between actors[86]. Some specific actions (and oppositions) are identified above, but further context-specific work is needed for each case and is a priority for future research. Here, three more general changes are identified that would enable positive tipping points for nature by affecting key 'leverage points' of systems change[87].

First, the process of social learning about positive tipping points for nature could be enhanced. Numerous bottom-up initiatives are already driving nature-positive change and opposing activities destroying nature. In so doing, groups of people are learning about what works and what does not work in a particular local context. That knowledge is transferable within an ecological and cultural context, and some of it more widely. Thus, enabling collective learning could accelerate the diffusion of positive tipping points for nature. To facilitate that, platforms that share learning in an informative and engaging way (for example, https://ecotippingpoints.org), can play a valuable role. A 'wiki' style 'Gaiapedia' that invites all those trying to drive positive tipping points for nature to share their knowledge and learning could help propel change—by affecting the leverage points of altering information flows and (social) system structure[87].

Second, properly valuing nature in economic decision-making could help shift investment to supporting positive tipping points. Even early steps to valuing ecosystem services, notably carbon sequestration, have enabled diffusion, for example, of agroforestry initiatives[68]. Most importantly, many life-supporting things nature does for us (such as photosynthesis) are not substitutable[88] with something else we can buy in the marketplace to provide the same services (or they are only partially substitutable). Recognizing this in calculations of 'utility' (collective wellbeing) immediately gives nature extra value, which tends to increase over time[89]. If there are also negative tipping points in nature where utility could be irreversibly lost, this further adds to the value of preserving nature now[90]. Means of factoring (un)substitutability into nature valuation have been proposed[91]. If economic actors had to pay the associated values when affecting nature, this could have considerable leverage through changing the rules of the (financial) system[87].

Finally, a more widespread adoption of eco-centric worldviews and associated legal status of nature would help enable other nature-positive tipping points. The dominant worldview in many societies remains an anthropocentric one in which nature is separate from

people and only there to serve human emancipation. The theory of mental coherence argues that people do not change their worldview incrementally. Rather, when cognitive dissonance reaches a critical level, individual worldviews can pass a tipping point[92]. There are already signs of a rise in ecocentrism in the form of the 'rights of nature' movement, with an increasing number of jurisdictions assigning legal status to non-human natural entities. Changing the ethical and legal status of nature is a powerful practical step to underpin nature-positive action. Such a tipping point in paradigm could be the deepest leverage point for nature-positive system change[87].

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Competing interests

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Additional information

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